

THE RELAXATION OF THE DOUBLE LAYER AROUND COLLOIDAL PARTICLES AND THE LOW-FREQUENCY DIELECTRIC DISPERSION

PART I. THEORETICAL CONSIDERATIONS

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ABSTRACT

This paper discusses the relaxation of electrical double layers around colloid particles and its role in low-frequency dielectric dispersion $\Delta\epsilon(\omega)$. As this dispersion is very sensitive to the distribution of ions over the various double-layer parts, relaxation measurements can provide information on the double-layer composition. A picture is elaborated in which the Schwarz theory is corrected for the contribution of polarization in the diffuse part of the double layer; the potential drop over this part under non-equilibrium conditions is computed from the polarization of the Stern layer. There are two contributions to $\Delta\epsilon(\omega)$, one stemming from bound counterions and one from concentration polarization in the diffuse part of the double layer. The conditions under which these two mechanisms are operative are discussed.

(I) INTRODUCTION

Over the past decades various models for the relaxation of the double layer around colloidal particles have been developed. Of these Overbeek's contribution in connection with his improved theory of electrophoresis [1] deserves special attention. Since the particle and the countercharge bear opposite charges, movement in opposite directions is effected by an applied outer electric field. Details of the hydrodynamics depend among other things on the position of the slipping plane, or rather on the mechanism of the slipping process. Owing to the two opposite movements, the original symmetry of the double layer is perturbed; conduction and diffusion processes try to restore the equilibrium distribution, but this restoration

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requires a certain time, the *relaxation time*, so that the ions in the outer double-layer part lag behind the particle. This leads to an additional electric field, directed opposite to the outer one, which reduces the electrophoretic mobility [2].

The most important characteristics of such a relaxation process are (1) the spatial deformation of the double layer, (2) the ensuing counteracting additional field (which we will designate as a “polarization field”) and (3) the important role of diffusion in the region beyond the double layer, the rate of which is coupled to the time of relaxation.

Double-layer relaxation has also been considered in connection with the theory of conductivity of suspensions [3,4], because current flow through double layers leads to their deformation. Recently, the important role of double-layer relaxation in the collision between charged particles (and hence its influence on colloid stability) has also been recognised [5,6]. It is mandatory to take double-layer relaxation into account if coagulation proceeds under non-equilibrium conditions [7], for instance under the influence of an applied electric field [8] or in a concentration gradient [9].

The relation between double-layer relaxation and these phenomena and the connection between this relaxation and the dielectric properties of suspensions [10–12] are related phenomena that should be treated from the same view point. For instance, it may be noted that the Dukhin–Shilov theory for the polarization of double layers [13,14] and its application to the dielectric properties of suspensions [10–12] and Stigter’s theory for the electric conductivity of suspensions [4], are both developed on the basis of Overbeek’s representation [1]. On the other hand, the Schwarz–Schurr theory [15,16] ignores the relaxation of the diffuse part of the double layer and is based on a quite different mechanism. The substantial discrepancies between the polarization mechanisms of Dukhin and Shilov [9–14] and that of Schwarz and Schurr [15,16] are due to different kinetic models of the double layer, and their respective qualities depend on the quality of the model assumptions.

It is the purpose of the present study to further our insight in this matter, combining new theoretical and experimental developments. In this first paper we discuss the present state of the theory, thereby proposing an improvement. We intend to follow this article with an experimental one (Part II) on model colloids (namely homodisperse polystyrene latices in various electrolytes), and one in which the modified theory is applied to the measurements (Part III).

(II) GENERAL CONSIDERATIONS. THE MODELS OF O’KONSKI AND SCHWARZ

In dilute suspensions, the contribution $\Delta\epsilon^*$ of dispersed particles to the complex dielectric permittivity ϵ^* in an alternating field of cyclic frequency ω can be shown to obey [11,17]

$$\Delta\epsilon^*/\epsilon_a^* = 4\pi n\mu^*/E \quad (1)$$

where n is the number of particles per unit volume, ϵ_a^* the complex dielectric permittivity of the dispersion medium and μ^* the effective induced dipole moment of the particle, i.e. the dipole moment divided by $4\pi\epsilon_0\epsilon_a$. It also occurs in the

expression for the potential ϕ_a far from the particles:

$$\phi_a = (-Er + \mu^*/r^2) \cos \theta \quad (2)$$

Furthermore,

$$E = E_0 e^{i\omega t} \quad (3)$$

is the field strength of the applied external uniform ac field, t the time and r and θ the spherical coordinates. The particle is assumed to be spherical, and θ is referred to the direction of the external field ($\theta = 0$).

Because of eqn. (1), those particle properties that are related to μ^* are also relevant for the dielectric permittivity of suspensions. Two of them deserve special attention. First, as a rule, the particle conductivity k_i and the dielectric permittivity ϵ_i of the particles differ from the corresponding quantities in the continuous medium, k_a and ϵ_a^* . The results of this are described in the Maxwell–Wagner theory [18]. According to this, the dispersion of $\Delta\epsilon^*$ is determined by the finite time required for the formation of a polarization charge.

The second important factor is the surface (excess) conductivity. In O’Konski’s theory this has been accounted for by introducing an integral surface conductivity [17]. The theory allows for free exchange of charges between the surface conductance layer and the bulk. Charge conservation along non-conducting particles at any angle θ with the direction of E is expressed through

$$\text{div}_s \kappa^\sigma E_s(\theta) + k_a E_n(\theta) = -\partial \sigma(\theta) / \partial t \quad (4)$$

where E_s and E_n are the tangential and normal component of the field strength near the particle surface respectively, κ^σ is the surface conductivity, and σ the surface charge density.

For a suspension of spheres, the O’Konski theory leads to expressions for $\Delta\epsilon(0)$ and the relaxation time that are formally equivalent to those of Maxwell–Wagner, provided in the latter k_i is replaced by $2\kappa^\sigma/a$, where a is the particle radius. The O’Konski model successfully explains $\Delta\epsilon^*(\omega)$ near the characteristic frequency of Maxwell–Wagner, but offers no satisfactory interpretation of the high values for the low-frequency dispersion of the dielectric permittivity that often occurs in suspensions [11,12].

An attempt to account for the low-frequency polarization was put forward by Schwarz [15]. This approach was not based on a specified model of the double layer. Schwarz attributed the double-layer polarization entirely to counterions bound to the spherical particle in a thin layer. Exchange between the bulk and the layer of surface conductivity was assumed to be absent. Schwarz’s charge conservation condition contains no normal flows and is written

$$\text{div}_s I_b(\theta) = -\partial \sigma_{bE}(\theta) / \partial t \quad (5)$$

* In order to avoid confusion with the reciprocal Debye length κ^{-1} , we shall use the symbol k for conductivities.

where I_b and σ_{bE} are the current and surface charge density (in the presence of the field E) respectively, in the layer of bound counterions. The charge σ_{bE} in eqn. (5) need not be identical to σ in eqn. (4). Because of the absence of exchange, I_b must contain a diffusion contribution. In dc fields ($\omega \rightarrow 0$), $I_b \rightarrow 0$; the surface current in the layer of bound ions vanishes because diffusion and conduction compensate for each other.

According to Schwarz, the expression for $I_b(\theta)$ contains a conduction (migration) and a diffusion part and can be written

$$I_b(\theta) = \frac{D_b}{a} \left[\frac{\partial \Delta \sigma_{bE}}{\partial \theta} + \frac{F \sigma_b}{RT} \frac{\partial \phi_{bE}}{\partial \theta} \right] \quad (6)$$

where $\Delta \sigma_{bE}$ is the deviation of the bound charge σ_{bE} at given θ and t in the presence of a field E from the same σ_b in the absence of a field, i.e.

$$\Delta \sigma_{bE}(\theta, t) = \sigma_{bE}(\theta, t) - \sigma_b \quad (7)$$

where D_b is the surface diffusion coefficient for bound ions and ϕ_{bE} is the electric potential in the layer of bound ions

$$\Delta \phi_{bE}(\theta, t) = \phi_{bE}(\theta, t) - \phi_b \quad (8)$$

In Schwarz's picture the time required to establish the distribution of bound charges in the stationary state is $a^2/2D_b$. The corresponding critical frequency of dielectric dispersion is

$$\omega_{b,cr} = 2D_b/a^2 \quad (9)$$

As stated above, the Schwarz model applies only in the absence of normal flow. As these flows are suppressed at high frequencies, the picture is applicable if

$$\omega \gg \tau_{bn}^{-1} \quad (10)$$

where τ_{bn} is the time required to establish equilibrium between the layer of bound ions and the bulk. Because of eqn. (9) the applicability range of the Schwarz theory can also be written

$$(1/\tau_{bn})(a^2/2D_b) \ll 1 \quad (11)$$

Condition (11) is not likely to be met in practice. First, eqn. (11) requires large values of D_b , but if the bound ions are so tightly associated with the surface that they do not exchange, this is unlikely. Second, the conditions do not concur with the kinetics of a diffuse double-layer part either. The relaxation time of a diffuse layer τ_d is on the order of

$$\tau_d \sim (2\kappa^2 D_d)^{-1} \quad (12)$$

where κ is the reciprocal Debye length and D_d the diffusion coefficient of ions in this layer (which may be lower than the corresponding value D in bulk). Schwarz developed his theory for thin layers, implying

$$\kappa a \gg 1 \quad (13)$$

so that, if the bound ions were distributed as in a diffuse double layer,

$$(1/\tau_d)(a^2/2D_d) \sim (\kappa a)^2 \gg 1 \quad (14)$$

which is in conflict with eqn. (11).

In addition to these objections, it may be repeated that Schwarz's double-layer theory is far too simple in that he only considers bound counterions. In line with the present view that the solution side of a double layer generally consists of two parts, a diffuse and a non-diffuse one, we shall generalize below the Schwarz theory by explicitly accounting for three categories of charges (Fig. 1):

- (1) fixed surface charges (e.g. covalently bound ionic groups), charge density σ_f ;
- (2) a layer of bound counterions (e.g. as ion pairs), charge density σ_b ;
- (3) a diffuse layer, charge density σ_d .

It would have been even more rigorous to subdivide the layer of bound ions into an inner and an outer Helmholtz layer, but we shall not pursue this refinement, partly because it would considerably complicate the mathematics and partly because the present picture already appears adequate to explain recent experimental results (Part III of this series).

The term "fixed charge" is interpreted as a charge with a negligible mobility along the surface. This is an important condition. With respect to counterions with surface diffusion coefficient $D_s < D$, for frequencies

$$\omega \gg 2D_s a^{-2} \quad (15)$$

they may be considered as fixed, whereas for

$$\omega \lesssim 2D_s a^{-2} \quad (16)$$

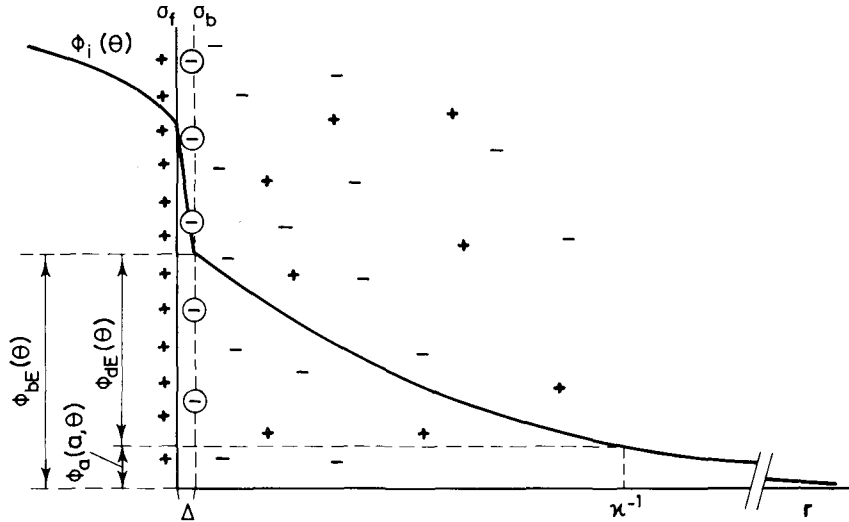


Fig. 1. The model for a polarized double layer used in the present study. Explanation in text. (At large distances ϕ equals Er because of eqn. 2.)

they are mobile, and if they do not exchange against the bulk they would be equivalent to the “bound ions” in Schwarz’s model. We recall that a Schwarz-type polarization layer *can* be realized if condition (11) applies.

In passing, we note that in this and following equations strictly $(a + \Delta)$ should be used instead of a , but as the difference is trivial we shall ignore it for simplicity.

If the ions in the inner Helmholtz plane are also fixed, the double-layer picture is again simple, with all ions outside the outer Helmholtz plane constituting the countercharge. However, if these ions are not considered as fixed, i.e. if eqn. (15) is not satisfied, the Schwarz model requires generalization. In that case the polarization of two layers of ions must be considered, with charge densities σ_b and σ_d .

(III) GENERALIZATION OF THE SCHWARZ MODEL

The application of an outer electric field leads to a shift in the distribution of bound charges, upon which the spherical symmetry is lost (Fig. 2a).

The diagram of Fig. 2a is the result of the tendency of all counterions to move in the direction opposite to that of the field E . It is typical for the bound ions that they

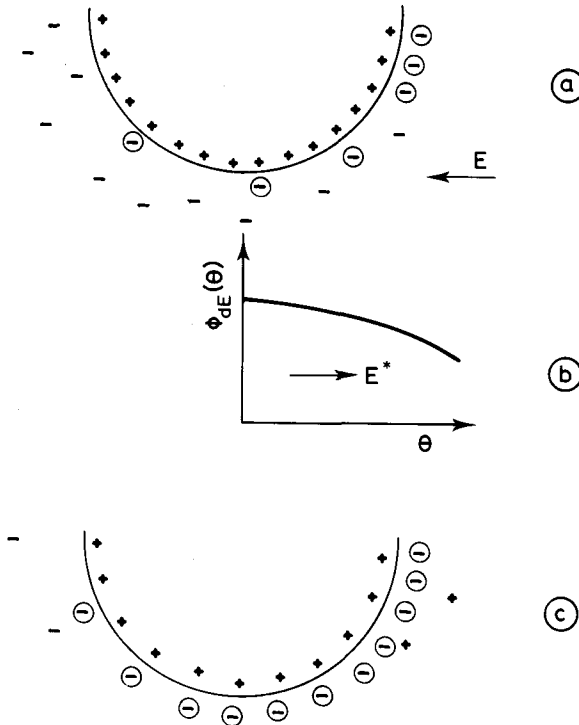


Fig. 2. (a) Illustration of polarization mechanism in a double layer; (b) illustration of the ensuing variation of $\phi_{dE}(\theta)$ with θ . The polarization field E^* is directed to the right; (c) polarization mechanism if $\sigma_d = 0$.

enrich the positive side (to the right in Fig. 2a), whereas diffuse ions can exchange with the bulk. To maintain (almost) electroneutrality at each θ , they enrich the negative side (to the left in Fig. 2a).

We assume that the applied field is uniform and harmonic (eqn. 3) and sufficiently weak, so that all quantities related to the field are linear with the field and vary as a function of time according to the same law as the field. Let us first consider the bound counterions, i.e. the only ions considered by Schwarz.

For σ_{bE} we have eqn. (7) with

$$\Delta\sigma_{be}(\theta, t) = \Delta\sigma_{bE}(\theta)e^{i\omega t} \quad (17)$$

Substituting eqn. (6) in eqn. (5), writing σ_b according to eqn. (7), $\Delta\sigma_{bE}$ following eqn. (17), and elaborating the second term on the rhs of eqn. (6) by using eqn. (8) and

$$\Delta\phi_{bE}(\theta, t) = \Delta\phi_{bE}(\theta)e^{i\omega t} \quad (18)$$

the following expression is obtained:

$$i\omega\Delta\sigma_{bE}(\theta) = \frac{D_b}{a^2} \frac{1}{\sin\theta} \frac{\partial}{\partial\theta} \left\{ \sin\theta \left[\frac{\partial\Delta\sigma_{bE}(\theta)}{\partial\theta} + \frac{F\sigma_b}{RT} \frac{\partial\phi_{bE}}{\partial\theta} \right] \right\} \quad (19)$$

which is identical to eqn. (15) in ref. 15. From this equation Schwarz obtained the following general form for the relationship between $\Delta\sigma_{bE}(\theta)$ and $\Delta\phi_{bE}(\theta)$:

$$\Delta\sigma_{bE}(\theta) = - \frac{F\sigma_b/RT}{1 + i\omega\tau_{bs}} \Delta\phi_{bE}(\theta) \quad (20)$$

with

$$\tau_{bs} = a^2/2D_b \quad (21)$$

The next step is to find the functions $\Delta\phi_{bE}(\theta)$ and $\Delta\sigma_{bE}(\theta, t)$, i.e. the oscillations of the bound and the diffuse ions with the outer field. The space beyond the boundaries of the double layer is electroneutral, so that according to Laplace

$$\nabla^2\phi_a(r, \theta) = 0 \quad (r > a + \kappa^{-1}) \quad (22)$$

which leads to eqn. (2).

As Schwarz only considers bound counterions, he identifies $\phi_{bE}(\theta)$ with $\phi_a(r, \theta)$ and substitutes the latter for the former in eqn. (19) and, hence, in eqn. (20). Let us discuss the result of this oversimplification, restricting ourselves to the most interesting low-frequency range, determined by $\omega\tau_{bs} \lesssim 1$. As in the Schwarz picture $\kappa a \gg 1$ (eqn. 13), τ_d in eqn. (12) is much smaller than τ_{bs} , so that $\omega\tau_d \ll \omega\tau_{bs} \lesssim 1$, implying that under these conditions redistribution of diffuse ions (if there is also a diffuse double-layer part) is sufficiently fast almost to retain equilibrium between the diffuse and the bound part of the double layer at any moment during polarization. Consequently, the surface and bound charges are continually almost completely screened by the diffuse part and hence the polarization of the diffuse layer cannot be ignored. In Appendix I the situation is rigorously analysed for small perturbations; there it will be proved that such perturbations may be neglected.

We conclude from the foregoing that the layer of bound charges is in fact continually screened by the diffuse charge, so that a more rigorous analysis is needed in which both the diffuse and the bound counterions are considered.

To achieve this, we write the surface charge distribution in the polarized diffuse part of the double layer, σ_{dE} , as the sum of the original symmetrical distribution σ_d , and a small excess $\Delta\sigma_{dE}(\theta, t)$ in a way similar to that for the bound ions eqns. (7) and (17)

$$\sigma_{dE}(\theta, t) = \sigma_d + \Delta\sigma_{dE}(\theta, t) \quad (23)$$

$$\Delta\sigma_{dE}(\theta, t) = \Delta\sigma_{dE}(\theta)e^{i\omega t} \quad (24)$$

The double layer as a whole is always approximately electroneutral at any θ

$$\sigma_f + \sigma_b(\theta) + \sigma_d(\theta) = 0 \quad (25)$$

$$\sigma_f + \sigma_{bE}(\theta, t) + \sigma_{dE}(\theta, t) = 0 \quad (26)$$

Using eqns. (7), (23) and (25), it follows from eqn. (26) that

$$\Delta\sigma_{dE}(\theta) = -\Delta\sigma_{bE}(\theta) \quad (27)$$

The Laplace equation in the form (22) does not hold for parts within the diffuse double layer, but it does apply for the potential distribution along the outer periphery of this layer, say, for potentials $\leq \phi_a(a + \kappa^{-1}, \theta)$. We shall identify the potential drop over the diffuse layer ϕ_{dE} as the drop between $r = a$, where $\phi = \phi_{bE}(\theta, t) = \phi_{bE}(a, \theta, t)$ and $r = a + \kappa^{-1}$ (Fig. 1b); hence

$$\phi_{bE}(\theta, t) = \phi_a(a + \kappa^{-1}, \theta, t) + \phi_{dE}(\theta, t) \quad (28)$$

in which by analogy to eqn. (8):

$$\phi_{dE}(\theta, t) = \phi_d + \Delta\phi_{dE}(\theta, t) \quad (29)$$

Strictly, the potentials should be referred to $\phi_a = 0$ attained at very high values of r (in principle, $r \rightarrow \infty$). However, in non-equilibrium double layers the difficulty arises that ϕ also keeps changing in the region of electroneutrality, so that the unambiguous establishment of a reference for ϕ becomes difficult. Therefore, we accept for these purposes $r = a + \kappa^{-1}$ as the outer double-layer boundary, considering the quasi-equilibrium of the part $r > a + \kappa^{-1}$.

Transformation of potential perturbations into charge perturbations proceeds by introducing differential capacitances. With eqn. (27)

$$\Delta\phi_{dE}(\theta) = -\Delta\sigma_{dE}(\theta)/C_d = \Delta\sigma_{bE}(\theta)/C_d \quad (30)$$

where

$$C_d = \epsilon_a \epsilon_0 \kappa \cosh(F\phi_d/2RT) \quad (31)$$

for a symmetrical electrolyte. Here, ϵ_a is the relative dielectric constant of the bulk solution. Strictly, for C_d we require only the capacitance of the part between $r = a$ and $r = a + \kappa^{-1}$, but substitution of the complete capacitance introduces only negligible error, since $\kappa a \gg 1$.

As illustrated by Fig. 2a, the quantities $|\sigma_{dE}(\theta)|$ and $|\Delta\sigma_{dE}(\theta)|$ decrease with increasing θ . Also, $\phi_{dE}(\theta)$ and $\Delta\phi_{dE}(\theta)$ decrease with θ . The polarization field E^* is directed towards decreasing ϕ_{dE} (Fig. 2b), i.e. opposite to the outer field E . This polarization field reduces the amplitude of the shifts of the bound counterions and hence the magnitudes of μ^* and $\Delta\epsilon^*$. The polarization of the diffuse part reduces the relaxation time of the bound ions. As Schwarz ignores the diffuse part, he overestimates the relaxation time and $\Delta\epsilon^*$. In passing, it may be added that this also applies for the case that $\sigma_d = 0$, because it is the oscillating part of the diffuse part ($\Delta\sigma_{dE}(\theta, t)$) that counts, and because of eqn. (27) this must persist if there is no net overall diffuse charge. Figure 2c illustrates this.

Differentiating eqn. (28) with respect to θ , using eqns. (30) and (27), gives

$$\frac{\partial\phi_{bE}}{\partial\theta} = \frac{\partial\phi_a}{\partial\theta} + \frac{\partial\phi_{dE}}{\partial\theta} = \frac{\partial\phi_a}{\partial\theta} - \frac{1}{C_d} \frac{\partial\Delta\sigma_{dE}}{\partial\theta} = \frac{\partial\phi_a}{\partial\theta} + \frac{1}{C_d} \frac{\partial\Delta\sigma_{bE}}{\partial\theta} \quad (32)$$

illustrating how the polarization of bound ions is influenced by the tangential oscillations of the diffuse atmosphere ($\partial\phi_{dE}/\partial\theta$). For very high diffuse capacitances the last term vanishes; in that case the angle dependence of ϕ_b is fully determined by that in ϕ_a .

Substitution of eqn. (32) in eqn. (19) leads to

$$i\omega\Delta\sigma_{bE}(\theta) = \frac{FD_bM}{a^2 \sin \theta} \frac{\partial}{\partial\theta} \left\{ \sin \theta \left[\frac{\partial\Delta\sigma_{bE}}{\partial\theta} + \frac{F\sigma_b}{RT} \frac{1}{M} \frac{\partial\phi_a}{\partial\theta} \right] \right\} \quad (33)$$

with

$$\begin{aligned} M &= 1 + \frac{F\sigma_b}{RTC_d} = 1 + \frac{\kappa\sigma_b}{2Fc \cosh(F\phi_d/2RT)} \\ &= \frac{2}{\cosh(F\phi_d/2RT)} \left[\frac{\sigma_f\kappa}{4Fc} - \sinh\left(\frac{F\phi_d}{2RT}\right) \right] \end{aligned} \quad (34)$$

where c is the electrolyte concentration entering the equation through $\kappa^2 = 2F^2c/\epsilon_a\epsilon_0RT$. We used eqns. (25) and (31) in the derivation. The factor M corrects the Schwarz equation for the contribution of the diffuse part of the double layer.

Equation (33) is an improvement of eqn. (19). The former follows from the latter by substituting D_bM for D_b and $\phi_a(\theta)/M$ for $\phi_{bE}(\theta)$. By a similar argument, one can derive

$$\Delta\sigma_{bE}(\theta) = - \frac{F\sigma_b/RT}{1 + i\omega\tau_{bs}} \frac{\phi_a(\theta)}{M} \quad (35)$$

and

$$\tau_{bs} = a^2/2D_bM \quad (36)$$

It is easily verified that these two expressions follow from eqns. (20) and (21) respectively, after application of the same substitutions as above.

In order to find the dipole moment we use a procedure equivalent to that applied

by Schwarz. Consider first the boundary condition for the conservation of free charge σ_{dE} in the absence of tangential transport in the diffuse layer, a situation concurring with the condition

$$R_{el} = k_d^\sigma / k_a a \ll 1 \quad (37)$$

which reads

$$k_a \frac{\partial \phi_a}{\partial r} \Big|_{r=a+\kappa^{-1}} = \frac{\partial \Delta \sigma_{dE}}{\partial t} = i\omega \Delta \sigma_{dE} \quad (38)$$

Let us designate the sum of the charge densities in all layers at a given θ as $\sigma_t(\theta)$. Under non-equilibrium conditions $\sigma_t(\theta) \neq 0$, implying that $\Delta \sigma_{dE}(\theta)$ and $\Delta \sigma_{bE}(\theta)$ vary in a different fashion tangentially to the surface, so that at any specific θ the double layer is unbalanced, although, of course, taken over the entire particle, electroneutrality prevails*. The non-equilibrium total charge gives rise to an electric field beyond its boundaries, or, for that matter, to a potential distribution. The two excess surface charges are connected through Gauss's theorem, applied to the double-layer part subtending θ , and can be expressed in terms of σ_t as follows (see also the note following eqn. A2 in Appendix A):

$$-\epsilon_0 \epsilon_i \frac{\partial \phi_i}{\partial r} \Big|_{r=a} + \epsilon_0 \epsilon_a \frac{\partial \phi_a}{\partial r} \Big|_{r=a+\kappa^{-1}} = -[\Delta \sigma_{bE}(\theta) + \Delta \sigma_{dE}(\theta)] = -\Delta \sigma_t(\theta) \quad (39)$$

Multiplication of eqn. (39) by $i\omega$ and adding the outcome to eqn. (38) yields

$$-K_i \frac{\partial \phi_i}{\partial r} \Big|_{r=a} + K_a \frac{\partial \phi_a}{\partial r} \Big|_{r=a+\kappa^{-1}} = -i\omega \Delta \sigma_{bE} \quad (40)$$

with $K_a = k_a + i\omega \epsilon_0 \epsilon_a$ and $K_i = i\omega \epsilon_0 \epsilon_i$. In this way we arrive at the boundary condition that was also used by Schwarz (eqn. 21 in ref. 15). Equation (40) does not contain σ_{dE} explicitly, but from the procedure to derive it it is clear that $\Delta \sigma_{dE}$ has been accounted for, i.e. that the screening of the bound charge density changes by the diffuse double-layer part is taken into account.

That Schwarz, although not using a specific double-layer representation, arrived at the same boundary condition (40) is due to the fact that he, because he ignored $\Delta \sigma_{dE}$, had to introduce into eqns. (38) and (39) the charge σ_{MW} , generated by Maxwell-Wagner polarization. This charge σ_{MW} is eliminated in a later stage because of the conditions corresponding to eqns. (39) and (40).

The equation for μ^* is obtained by substitution of $\phi_a(r, \theta)$ according to eqn. (2) and $\Delta \sigma_{bE}(\theta)$ according to eqn. (35) in eqn. (40). Solving the equation leads to

$$\text{Im } \mu^* = \frac{3}{2} \frac{k_b^\sigma}{k_a a} \frac{\omega \tau_{bs}}{1 + \omega^2 \tau_{bs}^2 (1 + k_b^\sigma / k_a a)^2} a^3 E \quad (41)$$

* In passing, it may be as well to point out that $\sigma_t(\theta) \ll \sigma_b$ (see also Appendix A). Hence, σ_t could be neglected in eqn. (26) but not in eqn. (39), because in the latter expression it governs the potential distribution sought.

where $k_b^0 = F^2 D_b |\sigma_b| / RT$. On the basis of eqns. (1) and (41) we obtain the following corrected Schwarz equation:

$$\Delta\epsilon(0)/\epsilon_a = \frac{9}{2} p \kappa a \frac{\sigma_f \kappa / 4Fc - \sinh(F\phi_d / 2RT)}{1 + \frac{2}{\cosh(F\phi_d / 2RT)} \left[\frac{\sigma_f \kappa}{4Fc} - \sinh\left(\frac{F\phi_d}{2RT}\right) \right]} \quad (42)$$

where p is the volume fraction of the dispersed material. This equation is also obtainable from eqn. (45) in ref. 15, replacing σ_f by $(\sigma_f - \sigma_d)/M$.

The contribution to $\Delta\epsilon(0)$ of the polarization of the bound ions at fixed σ_f depends strongly on the distribution of counterions over the Stern and the diffuse layer. It passes through a maximum as a function of σ_d because:

- (1) at high σ_d (i.e. $\sigma_d \rightarrow \sigma_f$) σ_b has to go down;
- (2) at low σ_d (i.e. $\sigma_b \rightarrow \sigma_f$) I_b goes down because of the opposing electric field, determined by the angular dependency $\Delta\phi_d(\theta)$; this latter quantity rises at low σ_d because of the low capacitance of the diffuse part.

It follows that the relative contribution to $\Delta\epsilon(0)$ of bound ions is very sensitive to the nature and concentration of the indifferent electrolyte and is a maximum around $\sigma_b \sim \sigma_d = \sigma_f/2$.

(IV) LOW-FREQUENCY DIELECTRIC DISPERSION, DETERMINED BY CONCENTRATION POLARIZATION OF A DIFFUSE ELECTRICAL DOUBLE LAYER

In section (III) the creation of μ^* due to bound ions was considered. In that case there were only normal flows in the diffuse part of the double layer, no tangential ones (eqn. 37). Now we consider the opposite case where Stern ions are absent ($\sigma_b = 0$) and polarization is only due to the diffuse part. It is typical for this new situation that electro-osmotic flow around the particle is to occur. Consequently, under these conditions we must introduce the notion of a slipping plane, delimiting the hydrodynamically movable and stationary parts of the liquid.

We shall assume that all ions are mobile, including those between the surface and the slipping plane, so that they contribute to the surface conductance, to μ^* and $\Delta\epsilon^*$. By way of approximation we shall not consider a detailed structure of this layer. Clearly it is bound by two equipotential surfaces. The outer surface is the slipping plane, characterized by the potential ζ . For the inner layers we choose the plane delimiting the fixed surface charge (including fixed counterions, if any). This is the Stern plane, characterized by ϕ_d . In other words, we now allow for the possibility that there is a layer in which the ions are mobile, even if the liquid is not. The assumption implies a possible difference between ϕ_d and ζ .

For frequencies $\omega \ll \tau_d^{-1}$ (eqn. 12) the diffuse layer is quasi-stationary, i.e. it can

* For a double layer at rest it is known that it expels electrolyte due to negative adsorption of *co-ions* (the Donnan effect). Under non-equilibrium conditions this distribution of neutral electrolyte around the particle becomes asymmetrical and it is due to the polarization of the *counterions*.

be described by the same equations that Overbeek used for the electrophoresis in a dc field [1]. It follows from Overbeek's theory that in an applied field the concentration of the neutral electrolyte beyond the limits of the double layer is changed *. In a dc field this change depends on the coordinate (as $\cos \theta/r^2$). The occurrence of a concentration drop in the outer field is very important for the low-frequency impedance of suspensions. According to refs. 10–12, 14 for $\omega \ll \tau_d^{-1}$ and $\kappa a \gg 1$ the diffuse double layer is not only quasi-stationary but is also in local equilibrium with the neutral electrolyte close to this part. This consideration permits an analytical approach to the relaxation in an ac field on the same basis as Overbeek [1], without making restrictions on the value of the potential drop over the diffuse layer [14].

According to refs. 10–12, the finite time to establish the concentration drop beyond the double-layer limits is the determining factor for the $\mu^*(\omega)$ dependence. This time is close to the time required to forward a diffusion front over a distance of the order of the particle radius ($\tau_c \sim a^2/2D$). In this case the induced dipole moment takes the following shape [10]:

$$\mu^* = -\frac{3}{2} E a^3 \frac{a_1 + a_1 W + i(a_1 W + a_2 W^2)}{A_1 + A_1 W + i(A_1 W + A_2 W^2)} \quad (43)$$

where

$$W = a\sqrt{\omega/2D} \quad (44)$$

$$a_1 = a^+(1 + \alpha_c^-) - \alpha^-(1 + \alpha_c^+) \quad (45)$$

$$A_1 = (z^- - \alpha^-)(1 + \alpha_c^+) + (z^+ + \alpha^+)(1 + \alpha_c^-) \quad (46)$$

$$a_2 = \alpha^+(1 + \alpha_1^-) - \alpha^-(1 + \alpha_1^+) \quad (47)$$

$$A_2 = (1 + \alpha_1^+)(z^- - \alpha^-) + (1 + \alpha_1^-)(z^+ + \alpha^+) \quad (48)$$

Here, z^+ and z^- are the valencies. Furthermore,

$$\alpha^\pm = \left[\frac{3}{2} m I_2^\pm \mp z^\pm I_1^\pm \right] / \kappa a \quad (49)$$

$$\alpha_c^\pm = \left[\frac{3}{2} m I_{2c}^\pm - I_1^\pm \right] / \kappa a \quad (50)$$

$$\alpha_1^\pm = z^\pm \left[I_1^\pm + I_3^\pm \right] / \kappa a \quad (51)$$

For symmetrical electrolytes * ($z^+ = z^- = z$)

$$I_1^\pm = -2 \left[\exp \left(\mp \frac{zF\phi_d}{RT} - 1 \right) \right] \quad (52)$$

$$I_2^\pm = \frac{2F\xi}{RT} \mp \frac{4}{z} \left[1 - \exp \left(\mp \frac{zF\xi}{2RT} \right) \right] \quad (53)$$

$$I_{2c}^\pm = \frac{2F\xi}{zRT} - \frac{4}{z^2} \left[1 - \exp \left(\mp \frac{zF\xi}{2RT} \right) \right] - 16 \ln \cosh \left(\frac{zF\xi}{4RT} \right) \quad (54)$$

* These and the following expressions apply for the simplifying condition $D^+ = D^- = D$. The more general case $D^+ \neq D^-$ can be found in ref. 12, p. 80ff. We intend to revert to this matter in part III.

$$I_3^\pm = - \frac{3 \cosh(zF\phi_d/2RT) - \exp(\mp zF\phi_d/RT) - 2}{\cosh(zF\phi_d/2RT)} \quad (55)$$

$$m = 2\epsilon_0\epsilon_a (RT/F)^2 / 3\eta D \quad (56)$$

We note that eqn. (52) is a conduction term, hence it contains ϕ_d ; on the other hand, eqns. (53) and (54) are electro-osmotic terms and therefore contain ζ .

The expression for the dielectric increment now reads

$$\frac{\Delta\epsilon(\omega)}{\epsilon_a} = \frac{9}{4} p (\kappa a)^2 \frac{(A_1 a_2 - A_2 a_1)(1 + W + W^2)}{(A_1 + A_1 W)^2 + (A_1 W + A_2 W^2)^2} \quad (57)$$

$$\frac{\Delta\epsilon(0)}{\epsilon_a} = \frac{9}{4} p (\kappa a)^2 \frac{A_1 a_2 - A_2 a_1}{A_1^2} \quad (58)$$

For sufficiently negatively charged particles, $I_1^+ \gg I_1^-$, $I_2^+ \gg I_2^-$, $I_{2c}^+ \gg I_{2c}^-$, and then eqn. (58) can be elaborated to

$$\frac{\Delta\epsilon(0)}{\epsilon_a} = \frac{9p(1+z)}{8} \left\{ \frac{\exp[zF|\phi_d|/2RT] + 3m/z^2 \exp[zF|\zeta|/2RT]}{1 + \frac{2 \exp[zF|\phi_d|/2RT]}{\kappa a} + \frac{6m \exp[zF|\zeta|/2RT]}{\kappa a z^2}} \right\}^2 \quad (59)$$

Thus, consideration of the change in the electrolyte concentration outside the double layer in an applied ac field enables us to formulate an expression for the low-frequency dielectric dispersion without having to introduce the notion of bound ions. However, this notion will be required if the Stern ions behave as if bound.

It follows from the considerations of sections (III) and (IV) that ion specificity in $\Delta\epsilon(0)$ can have two causes, depending on the question whether or not bound ions are present. If they are present, different counterions will bind to a differing extent and hence lead to different values of the ratio σ_d/σ_b . On the other hand, if they are absent, differences between D^+ and D^- (i.e. differences between m^+ and m^-) have to be invoked. In Part III of this series we intend to return to this matter.

(V) RESULTS AND COMPARISON OF POLARIZATION MECHANISMS

Some typical examples of dispersion curves are collected in Fig. 3a–d. These diagrams illustrate the points previously made, viz. that the low-frequency dispersion tends to be mainly due to polarization of the diffuse part of the double layer and that the contribution of bound counterions is a maximum around $\sigma_b \sim \sigma_f/2$. At given κa , the polarization of the purely diffuse double layer increases much more strongly with σ_f than polarization due to bound ions. At given σ_f there is a general increase with κa .

In Fig. 4 the boundary is given delimiting the areas where the contribution to the polarization of the diffuse double layer and the Stern layer respectively, prevail. To that end, $\Delta\epsilon(0)$ has been computed on the basis of eqns. (42) and (58) for the

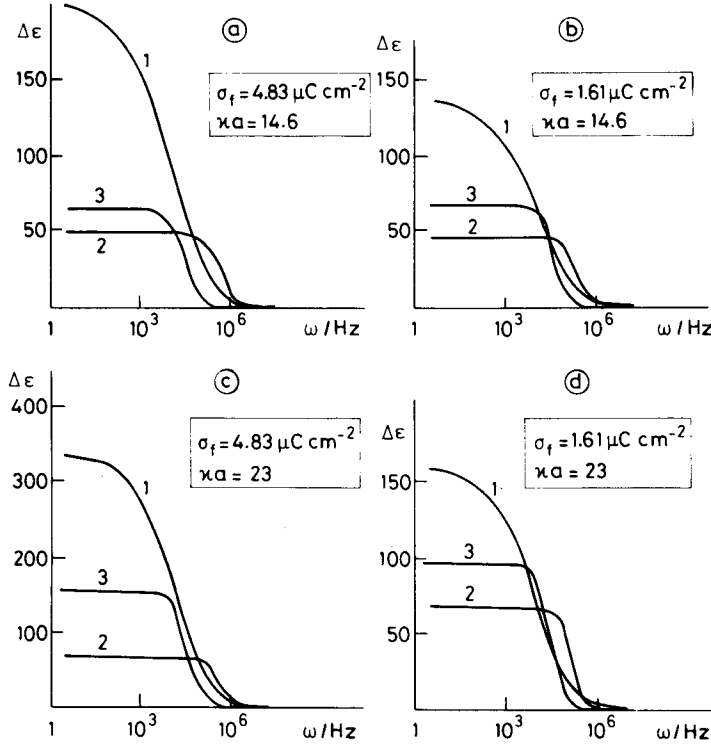


Fig. 3. Dielectric dispersion curves at different values of κa and the fixed charge σ_f . Volume fraction 0.02, $a = 0.22 \mu$, $D^+ = D^- = 2 \times 10^{-5} \text{ cm}^2 \text{ s}^{-1}$. Curve 1 polarization of diffuse atmosphere only (eqn. 57); curve 2: polarization due to bound counterions (improved Schwarz picture, eqn. (42); $\sigma_d = 0$; $\sigma_b = \sigma_f$; curve 3: as curve 2; $\sigma_d = \sigma_b = \sigma_f/2$. (Note the different ordinate axis scale in Fig. 3c.)

condition

$$\sigma_b = \sigma_d = \sigma_f/2 \quad (60)$$

at various σ_f and κa , taking $z = 1$. Curve 1 of this diagram is the relationship between σ_f and κa that, according to eqns. (42) and (58), leads to the same $\Delta\epsilon(0)$. Because of eqn. (60) this can also be written in terms of a relation between a certain critical value ϕ_d^{cr} and κa . Approximately

$$\sinh(zF\phi_d^{\text{cr}}/2RT) \sim 0.03 \kappa a \quad (61)$$

In the area left of this curve, the influence of the polarization of the diffuse layer prevails, whereas to the right of it an appreciable influence of the bound counterions must be expected (provided eqn. 13 is valid). Hence, the influence of bound counterions plays an important role only at moderate values of $\phi_d < \phi_d^{\text{cr}}$. Then $\Delta\epsilon(0)$ is much lower than its maximally attainable value attained at the same κa if $\phi_d \rightarrow \infty$ (this maximum is then determined by concentration polarization). To estimate the

maximum value of $\Delta\epsilon(0)$ at fixed σ_f and given κa , as determined by the bound counterions, computations on the basis of eqn. (42) have been carried out for the combination of σ_f and κa values satisfying curve 1. Results are given in curve 2.

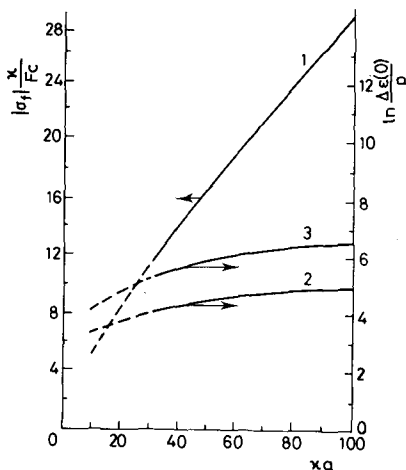


Fig. 4. Curve 1 represents pairs of values σ_f , κa for which $\epsilon(0)$ calculated according to eqns. (58) and (42) is the same, assuming $\sigma_d = \sigma_b = \sigma_f/2$. In the (σ_f, a) range left of the curve, eqn. (58) gives a higher value for $\Delta\epsilon(0)$ than eqn. (42); to the right of the curve it is the other way round. Curve 2 gives the pertaining $\Delta\epsilon(0)$ per unit volume fraction p according to eqn. (42), and curve 3 is the maximally attainable value of $\Delta\epsilon(0)$ according to eqn. (58). Below $\kappa a \sim 30$, the theory is only approximately valid.

Curve 3 characterizes the maximum value if $\Delta\epsilon(0)$ is only determined by concentration polarization according to eqn. (59) with $\exp(F\phi_d/2RT) \gg \kappa a$.

The extent of counterion binding can be described by a Stern-type equation [19] that we shall use in the simplified form

$$\frac{\sigma_b}{\sigma_f} = - \frac{\beta x \exp(-F\phi_d/RT)}{1 + \beta x \exp(-F\phi_d/RT)} \quad (62)$$

where β is the binding constant ($= \exp(-F\Phi_i/RT)$ if Φ_i is the specific adsorption potential), $x = 0.018 c$, with c in mol l^{-1} . An equal distribution (eqn. 60) is obtained if $\beta x \exp(-F\phi_d/2RT) = 1$ or, considering eqn. (61), if $\beta \sim x^{-1} \sim 50/c$. Complications due to electrode polarization limit measurements of the low-frequency dielectric dispersion to concentrations above $10^{-2} \text{ mol l}^{-1}$. Hence, it follows that a strong influence of bound counterions (relative to that of polarization of the diffuse part of the double layer) is possibly only at high values of β , of the order of 10^4 . As in most systems (AgI, Hg, lattices) $F\Phi_i$ is only a few kT , such high β -values are not usually attained except in the case of strongly specifically adsorbing counterions.

From all of this it follows that under most experimental situations $\Delta\epsilon(0)$ is determined by the concentration polarization mechanism.

(VI) COMMENTS ON OTHER TREATMENTS

Schurr [16] has proposed some modifications of the Schwarz model, but the modified picture is still far from perfect. He distinguishes two types of surface conductivity, one frequency-dependent and due to bound charges, and one frequency-independent contribution. This second contribution resembles the O'Konski treatment [17] and Henry's approach to electrophoresis [20]. However, the diffusion current of ions in the diffuse layer is ignored, as it is in the Schwarz theory, and this implies that the tangential electric field at its inner and outer boundary are identified. This means that Schurr's model is internally inconsistent, because it does take into account electric conduction in the diffuse layer, but not the ensuing concentration drop beyond the boundaries of the double layer.

The simultaneous accounting for the polarization of the layer of bound ions and that of the diffuse part is a rather elaborate and as yet unsolved problem. However, in the special case of low k_b^g and low k_d^g (both $\ll 2k_a a$), there is no mutual influence of the tangential flows in the bound and diffuse layer part, allowing the relaxations of these layers to be treated independently. In that case, $\Delta\epsilon^*(\omega)$ is simply a superposition of the contributions of these two layers.

If condition $k_d^g \ll 2k_a a$ is not fulfilled, the radial current in eqn. (40) is not only determined by the tangential flow of the bound counterions but also by that of the diffuse-layer ions. Schurr took this into account by the introduction in eqn. (40) of the extra term

$$\frac{k_d^g}{a^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \phi_{bE}}{\partial \theta} \right) \quad (63)$$

Solving eqn. (40) after this addition leads to his formula for $\Delta\epsilon(0)$. However, as follows from estimation (61), the condition $k_d^g \ll 2k_a a$ is very well obeyed for values of κa and σ_f , for which the effect of bound counterions on $\Delta\epsilon(0)$ is not small in comparison with the contribution of concentration polarization of the diffuse double-layer part. This condition breaks down only for substantially higher values of σ_d and ϕ_d when concentration polarization prevails, so that the addition of eqn. (63) is ineffective.

Ballario et al. [21] applied the Schwarz-Schurr theory to their data in solutions of very low c , but such a procedure is defective because it demands the use of Bikerman's surface conductivity equation in eqn. (63) which is allowed only if eqn. (13) is satisfied and this, in turn, is not the case at very low c . In a later paper [22] dealing with not so extremely low c , in an attempt to account for discrepancies with the Schwarz-Schurr theory, a phenomenologically adjustable parameter was introduced to account for the distribution of relaxation times. No physical meaning could be assigned to this parameter.

The theory for the polarization of the diffuse double layer by Dukhin and Shilov recently found a sequel in publications by O'Brien, White and DeLacey [23,24]. The physical picture of the two theories is identical, both regarding electrophoresis and dielectric phenomena, although O'Brien used only numerical methods. Both use the

law of the conservation of ions of every type, Poisson's law, and hydrodynamic equations. However, a difference emerges in that Dukhin and Shilov in their dielectric theory do not consider electrophoretic oscillations of the particles in ac fields. In addition, the O'Brien–White theory for dielectric dispersion is not restricted to large κa . This difference, however, does not lead to differences in the outcome, because at high κa , which is the applicability range of the Dukhin–Shilov theory, the contribution to μ^* by the double-layer deformation due to electrophoretic oscillations is much smaller than the contribution following from the deformation due to the applied field E (Appendix B). This has already been demonstrated in a theory dealing with the contribution of electrophoretic oscillations to the conductivity of suspensions [25].

Systematic comparison of the two theories is thwarted by the fact that the O'Brien–White theory has a numerical character; these authors present curves for the real and imaginary part of the dielectric permittivity only for one single large κa value, viz. $\kappa a = 50$ (Figs. 2 and 7 in ref. 24). In any case, the curves computed by our eqn. (57) for $\kappa a = 50$, $\phi_d = 50$ mV, $z = 1$, are very close. Three parameters, namely τ , $\epsilon(0)$ and $\epsilon''_{\max}$ can act as characteristic quantities. For $\kappa a = 50$, these parameters are identical in the two theories. It is important that for fixed a and D for $\kappa a = 50$ and 10, τ is the same in the two approaches and independent of ϕ_d . Also, the frequency at which ϵ'' attains its maximum value does not depend on κa , nor does it on ϕ_d . Inspection of Figs. 1, 2, 7, 8 or O'Brien and White [23] leads to the conclusion that for $\kappa a = 10$ or 50 and various values of ϕ_d the relation

$$\epsilon''_{\max} = 0.28 \epsilon(0) \quad (64)$$

is valid. This equation also follows from the Dukhin–Shilov theory as was demonstrated by Springer [26].

For lower values of κa , but still $\gg 1$, the discrepancies between the two theories grow with respect to $\Delta\epsilon(0)$ but not with the other characteristics.

Summarizing, the work by O'Brien, White and DeLacey [23,24], if made more accessible, would provide a picture over a wide κa -range provided relaxation is determined by the diffuse layer only. However, as the role of tangential flow in the dense layer or hydrodynamically stagnant part of the diffuse double layer is underestimated, this approach is as yet less suitable to exploit dielectric and related studies on colloidal systems to investigate the double-layer structure.

(VII) CONCLUSION

In this paper it was demonstrated that $\Delta\epsilon(\omega)$ in the Schwarz and Schurr theory is overestimated, because these authors neglect the fact that in order to maintain near-electroneutrality of the double layer as a whole at any section between θ and $\theta + \Delta\theta$, the diffuse part of the double layer becomes polarized in a direction opposite to that of the applied field. The associated polarization field acts as a brake on the flow of bound ions and, hence, reduces their contribution to the polarization. Incorporation of the diffuse double-layer contribution into the theory not only leads

to lower values of $\Delta\epsilon(0)$ but also renders the relaxation time dependent on double-layer parameters.

The influence of bound (i.e. Stern layer) ions, if any, on $\Delta\epsilon(\omega)$ is felt only in a restricted area of ϕ_d and κa ; only below a certain critical value of ϕ_d do bound ions contribute; above this value polarization is governed solely by the diffuse part. The conclusions on the dominating role of the diffuse-layer polarization was also independently obtained by DeLacey and White.

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APPENDIX A

From eqn. (39) it follows that

$$\Delta\sigma_{dE}(\theta) = -\Delta\sigma_{bE}(\theta) - \epsilon_0 \left(\epsilon_a \frac{\partial\phi_a}{\partial r} - \epsilon_i \frac{\partial\phi_i}{\partial r} \right) \quad (A1)$$

The second term in the rhs can be estimated by writing the condition for potential continuity upon transfer through the layer of bound ions (henceforth we shall assume that there are no free charge carriers inside the particle, i.e. $K_i = 0$).

$$\phi_i|_{r=a} = \phi_a|_{r=a} + \Delta\phi_{dE}(\theta) \quad (A2)$$

In this expression the potential drop $\phi_i(\theta) - \Delta\phi_{bE}(\theta)$ contribution of the inner part has been neglected because there is no screening of the charge due to bound ions by the particle side of the double layer. In other words, for the purpose of the description of the non-equilibrium situation, variations in ϕ_{bE} can be ignored since $|(\phi_i - \phi_{bE})/\phi_i| \sim \Delta/a \ll 1$.

Furthermore, we can write

$$\phi_i = \frac{\phi(a) + \Delta\phi_{dE}(\theta)}{a} \cdot r \cos \theta \quad (A3)$$

$$\phi_a = -Er \cos \theta + \mu^* \cos \theta / r^2 \quad (A4)$$

From eqn. (A2) it follows that

$$\frac{\partial\phi_i}{\partial r}|_{r=a} = \frac{\phi_a + \Delta\phi_{dE}(\theta)}{a} \quad (A5)$$

and from eqn. (A4),

$$\frac{\partial\phi_a}{\partial r}|_{r=a} = -\frac{2}{a}\phi_a(a) - 3E \cos \theta \quad (A6)$$

Substituting eqns. (A5) and (A6) in eqn. (A1), using eqn. (31), we obtain

$$\Delta\phi_{dE}(\theta) = \frac{\Delta\sigma_{bE}(\theta)}{C} - \frac{1}{\kappa a \cosh(F\phi_d/2RT)} \left\{ \frac{\epsilon_i \phi_a(a)}{\epsilon_a} + 2\phi_a(a) - 3Ea + \frac{\epsilon_i}{\epsilon_a} \Delta\phi_{dE}(\theta) \right\} \quad (A7)$$

The last term on the rhs of this expression may be neglected against the lhs because $\kappa a \gg 1$ and $\epsilon_i < \epsilon_a$ for aqueous systems. For the same reasons it is also allowed to neglect the first term in brackets against the second.

Under condition $2\kappa_b^0 \ll \kappa a$ and very low frequency $\omega \ll a^2/2D_b$, the current in the layer of bound ions is virtually zero (as it is in a constant field), so that for a non-conducting particle

$$\phi_a = -Er \cos \theta - Ea^3 \cos \theta / 2r^2$$

so that

$$\phi_a(a) - 3Ea/2 \quad (A8)$$

At sufficiently high frequencies, if $(2D_b\kappa^2)^{-1} \gg \omega \gg a^2/2D_b$, the diffusional counterflow is irrelevant so that the contribution of the conductivity of bound ions becomes more prominent. In that case

$$\phi_a = -Er \cos \theta + \frac{2k_b^0/a - k_a}{2k_b^0/a + 2k_a} \cdot \frac{a^3}{r^2} E \cos \theta$$

$$\phi_a(a) = -\frac{3Ea}{2 + 2k_b^0/k_a a} \quad (A9)$$

The variation of ϕ_a with ω is given by a Debye-type curve, therefore at $\omega \simeq \tau^{-1}$ one may write

$$\phi_a(a)|_{\omega=\tau^{-1}} \simeq \phi_a(a)|_{\omega=0} - \frac{1}{2} \langle \phi_a(a)|_{\omega=0} - \phi_a(a)|_{\omega=\infty} \rangle \quad (A10)$$

Substitution of eqns. (A8) and (A9) in eqn. (A10) leads to

$$\phi_a(a)|_{\omega=\tau^{-1}} \simeq -\frac{3Ea}{2} \left\{ 1 - \frac{k_b^0/2k_a a}{1 + k_b^0/k_a a} \right\} \quad (A11)$$

Using (A11) it is easy to see that

$$|2\phi_a(a) + 3Ea|_{\omega=\tau^{-1}} \leq |\phi_a(a)|_{\omega=\tau^{-1}} \quad (A12)$$

for all k_b^0 , provided

$$0 < \omega \leq 2D_b/a^2 \quad (A13)$$

Hence, upon substitution of eqn. (A7) in eqn. (28), it is again permissible to neglect $[2\phi_a(a) + 3Ea]/\kappa a \cosh(F\phi_d/2RT)$ against $\phi_a(a)$, so that

$$\Delta\phi_{dE}(\theta) \simeq \Delta\sigma_{bE}(\theta)/C_d \quad (A14)$$

implying that for the screening of the bound charges it is not necessary to account for the (small) deviations from the equilibrium situation.

APPENDIX B

That electrophoretic oscillations do not significantly contribute to $\Delta\epsilon(\omega)$ can also be understood on the basis of the following argument.

Motion of the particle brings about a double-layer deformation, creating an additional dipole moment μ_v^* , proportional to the electrophoretic mobility u , which in turn can be estimated from Smoluchowski's corresponding expression for the sedimentation potential [24]

$$\mu_v^* = \frac{3\epsilon\epsilon_0\zeta a}{2k_a} u = \frac{3}{2} \frac{a(\epsilon\epsilon_0\zeta)^2}{2\eta k_a} E^* \quad (B1)$$

The value of μ_v^* can be estimated by considering its influence on the electric conductivity k_{susp} of suspensions, which is related to $Re\mu_v^*$. In ref. 24 it was proven that if eqn. (13) is valid, μ_v^* is smaller than μ^* , the contribution to μ^* from the direct action of E on the double layer. Next, it was also ascertained that electrophoretic oscillations do not influence k_{susp} . Dielectric phenomena being connected with Im^* , the same reasoning can be applied to assess the influence of μ_v^* . Because of eqn. (2),

$$Im E^*(\theta) \simeq Im(\mu^*/a^3) \quad (B2)$$

Substitution of $Im(E^*)$ in eqn. (B1) instead of E^* allows an estimation of the ratio

$$\frac{Im \mu_v^*}{Im \mu^*} \sim \frac{3a(\epsilon\epsilon_0\zeta)^2}{2\eta k_a a^3} \quad (B3)$$

which, by using $k_a = \epsilon\epsilon_0 D\kappa^2$, $m = \epsilon\epsilon_0(\kappa T)^2/\eta De^2$, can be rewritten

$$\frac{Im \mu_v^*}{Im \mu^*} = \frac{3m}{2(\kappa a)^2} \left(\frac{e\zeta}{kT} \right)^2 \quad (B4)$$

As $m \sim 0.2$, it follows that for $\kappa a \gg 1$ electrophoretic oscillations can be neglected, also for dielectric phenomena, since they are a second-order feature.

REFERENCES

- 1 J.Th.G. Overbeek, *Kolloid Beih.*, 54 (1943) 287.
- 2 J.Th.G. Overbeek, *Adv. Colloid Sci.*, 3 (1950) 97.
- 3 S.S. Dukhin, *Croat. Chem. Acta*, 48 (1976) 574.
- 4 D. Stigter, *J. Phys. Chem.*, 82 (1978) 1424.
- 5 J.Th.G. Overbeek, *J. Colloid Interface Sci.*, 58 (1977) 408.
- 6 J. Lyklema, *Pure Appl. Chem.*, 52 (1980) 1221; J. Lyklema and H.P. van Leeuwen, *Adv. Colloid Interface Sci.*, 16 (1982) 127.
- 7 S.S. Dukhin, *Croat. Chem. Acta*, 52 (1980) 167.
- 8 V.N. Shilov and V.R. Estrela-Lopis, in *Ustoichivost tonkikh plenok*, Nauka, Moscow, 1979, p. 35.
- 9 S.S. Dukhin in B.V. Deryagin (Ed.), *Issledovaniya b oblasti poverkhnostnykh sil*, Academy of Sciences, Moscow, 1961 p. 738 (trans. by Consultants Bureau, N.Y., as *Research in Surface Forces*, Vol. I, 1963, p. 27).

- 10 V.N. Shilov and S.S. Dukhin, *Kolloid Zh.*, 32 (1970) 293.
- 11 S.S. Dukhin in E. Matijević (Ed.), *Surface and Colloid Science Vol. 3*, Wiley-Interscience, New York, p. 293.
- 12 S.S. Dukhin and V.N. Shilov, *Dielectric Phenomena and the Double Layer in Disperse Systems and Polyelectrolytes*, Wiley-Interscience, New York, 1974.
- 13 S.S. Dukhin in B.V. Deryagin (Ed.), *Issledovaniya b oblasti poverkhnostnykh sil*, Academy of Sciences, Moscow, 1967, p. 335 (trans. by Consultants Bureau, NY, as *Research in Surface Forces*, Vol. III, 1971, p. 287).
- 14 V.N. Shilov and S.S. Dukhin, *Kolloid Zh.*, 32 (1970) 117.
- 15 G. Schwarz, *J. Phys. Chem.*, 66 (1962) 2636.
- 16 J.M. Schurr, *J. Phys. Chem.*, 68 (1964) 2407.
- 17 C.T. O'Konski, *J. Phys. Chem.*, 64 (1960) 605.
- 18 J.C. Maxwell, *A Treatise on Electricity and Magnetism*, Clarendon Press, Oxford, 1881.
- 19 O. Stern, *Z. Elektrochem.*, 30 (1924) 508.
- 20 D.C. Henry, *Trans. Faraday Soc.*, 44 (1948) 955.
- 21 C. Ballario, A. Bonincontro and C. Cametti, *J. Colloid Interface Sci.*, 54 (1976) 415.
- 22 C. Ballario, A. Bonincontro and C. Cametti, *J. Colloid Interface Sci.*, 72 (1979) 304.
- 23 R.W. O'Brien and L.R. White, *J. Chem. Soc. Faraday Trans. II*, 74 (1978) 1607.
- 24 E.H.B. DeLacey and L.R. White, *J. Chem. Soc. Faraday Trans. II*, 77 (1981) 2007.
- 25 S.S. Dukhin and B.V. Deryagin in E. Matijević (Ed.), *Surface and Colloid Science*, Vol. 7, Wiley, 1974, Ch. 2, especially eqn. (226) ff.
- 26 M.M. Springer, Thesis, Agricultural University, Wageningen, The Netherlands, 1979.